TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 2

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- Read each question **carefully**
- Answer **all** questions
- Write your answers **clearly** in the spaces provided
- **Check** your answers.

Information:

- This examination has **two** sections. These are clearly labelled
- You are **not allowed** to use a calculator for Section A
- You **are allowed** to use a basic calculator for Section B
- The **maximum** mark for this examination is 60
- The marks available for each question are shown in **bold** beneath each question.

exam paper

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

- Assessment: Past Paper 2 (Non-Calculator & Calculator)
- Complete stepped calculations, methods & mark scheme codes [M1/A1]
- Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)
- Pass standard: 36 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Section A	15		
Section B	45		
Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 9.028 – 0.17

Calculate: 14.508 - 5.65 = 8.858

Answer	8.858 ✓ [1 mark]

[1]

2. Work out 24.108 ÷ 4

Calculate: 18.081 ÷ 3 = 6.027

Answer	6.027 ✓ [1 mark]

[1]

Past paper

Questions continue on the following page

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3. 50 people are asked what pets they own.

The results are shown below:

	Rabbit	Cat
Dog	16	24
Fish	2	8

Step 1: P(outcome) = 24 / 50 = 48 / 100

One person is chosen at random.

Convert to decimal: 48 ÷ 100 = 0.48

What is the probability that the person chosen has a dog and a cat?

Give your answer as a decimal.

0.48 ✓ *[2 marks]*

Answer

Formula: Density = Mass ÷ Volume

Density = 1800 g ÷ 450 cm³ = 4 g/cm³

[2]

4. The mass of a sphere is 1800 grams (g).

The volume of the sphere is 450 cm^3

4 g/cm ✓ *[2 marks]*

Work out the density of the sphere.

Answer

[2]

Past paper

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5. Jesse is the manager of an aquarium.

The table below shows how many people visited the aquarium over 3 years:

Year	Number of people
2019	568 750
2020	128 528
2021	482 935

Step 1: Sum attendance: 568 750 + 128 528 + 482 935 = 1 180 213

Jesse thinks that more than 1 200 000 people visited the aquarium during the 3 years.

Comparison: 1 180 213 > 1 150 000. Therefore No.

Is Jesse correct?

Show how you decide.

No (1 180 213 < 1 200 000) [2 marks]

Answer	

[2]

6. Work out 1.324 + 2.206 + 14.75

Answer	

[2]

Questions continue on the following page

Past paper

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7. Work out:

$$\frac{3}{20} + \frac{17}{15}$$

Give your answer as an improper fraction.

1.324 + 2.206 + 14.75 = 18.28

18.28 ✓ [2 marks]

Answer	

[2]

8. Riley's annual salary is £32 000 before tax.

Riley knows that:

Common denominator: 3/20 = 9/60 and 17/15 = 68/60

- £12 570 will be tax free
- 20% tax will be deducted from the amount above £12 570

9/60 + 68/60 = 77/60 (or 1 17/60)

How much will the annual salary be after tax has been deducted?

77 / 6 ✓ [2 marks]

Answer	£

[3]

End of Section A.

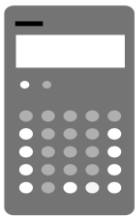
Past paper

Section B begins on Page 6.

Past paper

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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

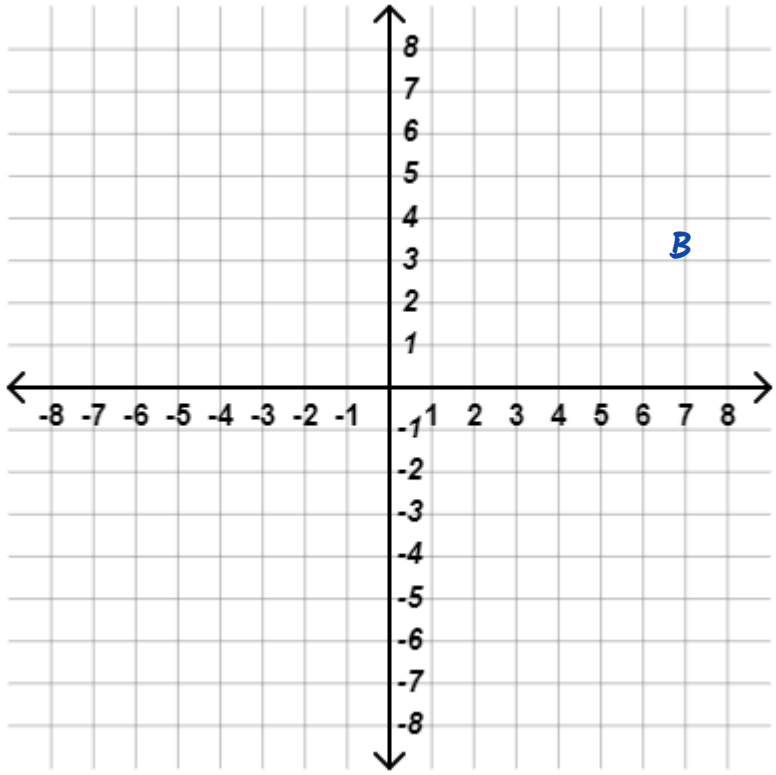
1. Write 13 000 000 000 in words.

Thirteen bil ✓ [1 mark]

Answer	

[1]

2. Plot the point (6, –3) on this coordinate grid:



B ✓ [1 mark]

[1]

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3. At football training, a footballer has 56 shots at goal and scores 42 times.

The football coach thinks that the footballer scored $\frac{7}{8}$ of the shots.

Has the coach worked out the correct fraction?

Show how you decide.

42 ÷ 56 = 3/4 = 6/8. (7/8 would be 49).

No (fraction 6/8) [2 marks]

Answer	

[2]

4. Work out:

BIDMAS calculation: 0.4

$$\left(\frac{25 + 5 \times 3}{10^2}\right)$$

Give your answer as a decimal.

0.4 [2 marks]

Answer	

[2]

Questions continue on the following page

Past paper

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5. 3480 people visited a museum in July.

5150 people visited the same museum in August.

Calculate the percentage change in the number of people from July to August.

Percentage increase = ((5150 - 3480) ÷ 3480) × 100 = 48% (47.99%)

Answer	48% ✓ [2 marks]	%

[2]

6. Leslie is planning a road trip.

The scale on the map is 1 : 75 000

The distance Leslie wants to travel represents 11 centimetres (cm) on the map.

Map scale: 11 cm x 75 000 = 825 000 cm = 8.25 km. No.

Leslie calculates this to be 82.5 kilometres (km).

Is Leslie correct?

Show how you decide.

No (8.25 km > 8 ✓ [2 marks]

Answer	

[2]

Past paper

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7. Nicky's homemade pillowcases are made using a piece of fabric two metres square.

Each time a pillowcase is made, 0.135 square metres of fabric is wasted.

Nicky wants to **estimate** what percentage of the total fabric is wasted.

Nicky rounds 0.135 to one decimal place.

Using the rounded value, what percentage should Nicky get?

Show your working.

Calculation: (0.1 ÷ 2) × 100 = 5%

5% ✓ [3 marks]

Answer	%

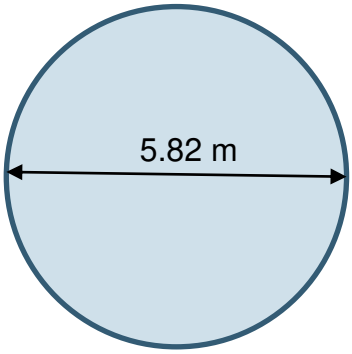
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Past paper

Questions continue on the following page

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8. Charlie wants to put a safety rope around this pond:



Not drawn to scale

Charlie thinks the total length of rope needed will be more than 18.5 metres (m).

Is Charlie correct?

Show how you decide.

Use $\pi = 3.14$

Circumference = π × d = 3.14 × 5.82 = 18.27 m. 18.27 < 18.5 m.

No (18.27 m < 18.5 m) ✓ [3 marks]

Answer	

[3]

Past paper

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9. Harper paid £12 500 into a new savings account 2 years ago.

The savings account paid 3% compound interest per year.

Harper wants to buy a new car for £18 750

Harper puts all the money in the savings account towards the new car.

How much more money is needed to pay for the car?

Compound interest: £12 500 × $(1.03)^2$ = £13 261.25

Remaining balance = £18 750 − £13 261.25 = £5488.75

£5488 ✓ [3 marks]

Answer	£

[3]

Past paper

Questions continue on the following page

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10. Alex needs to know what time to leave home to get to a dentist appointment by 10:40 am.

Alex can cycle at an average speed of 12 miles per hour.

The dental surgery is 15 miles from Alex's house.

What is the **latest** time Alex can leave home to cycle to the appointment?

Travel time = 15 km ÷ 12 kph = 1.25 hours = 1 hour 15 minutes

Arrival time = 8:10 am + 1 hr 15 mins = 9:25 am

Answer	**am**

9:25 a ✓ [3 marks]

[3]

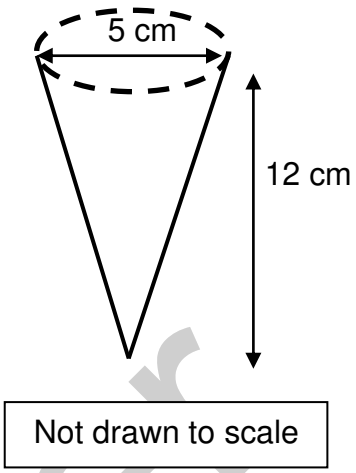
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11. Taylor is going to fill a cone with strawberry and vanilla ice-creams.

Taylor will use strawberry and vanilla in the ratio 2 : 3

How much of each flavour will Taylor need?



Not drawn to scale

Use:

$$v = r^2 h$$

Where:
v = volume of cone
r = radius of cone
h = height of cone

Volume = 2.5 × 2.5 × 12 = 75 cm³. Ratio 2 : 3 (5 parts)

Strawberry = 75 ÷ 5 × 2 = 30 cm³ | Vanilla = 75 ÷ 5 × 3 = 45 cm³

Strawberry	30 ✓	**cm³**
Vanilla	45 ✓	**cm³**

[4 marks]

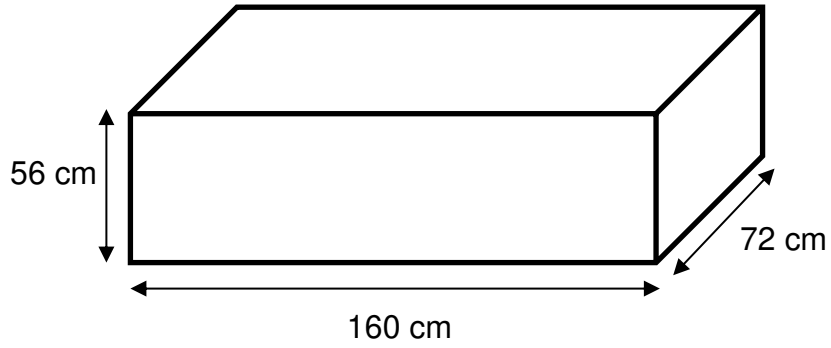
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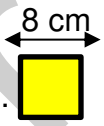
Questions continue on the following page

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12. Jamie needs to cover this cuboid chest in multi-coloured stickers for an art project:



Not drawn to scale



The stickers are in the shape of a square with a side length of 8 cm.

Total Surface Area = 2[(160×56) + (160×72) + (72×56)] = 49 024 cm²

To work out how many stickers are needed, Jamie uses this formula:

Each sticker = 8 × 8 = 64 cm² --> Stickers = 49 024 ÷ 64 = 766 stickers

$$\text{Total number of stickers} = \frac{\text{Total surface area of chest}}{\text{Area of one sticker}}$$

How many stickers does Jamie need?

766 stickers [4 marks]

Answer	

[4]

Past paper

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13. Drew is a gardener.

This table shows how much other local gardeners charge per hour:

Amount (A) charged per hour (£)	Number of gardeners
20 < A ≤ 30	2
30 < A ≤ 40	6
40 < A ≤ 50	8
50 < A ≤ 60	4
Total	20

Midpoints: 25, 35, 45, 55. Total fx = 50 + 210 + 360 + 220 = 840

Total hours = 20 --> Estimated Mean = 840 ÷ 20 = £42 per hour

Drew wants to charge an hourly rate equal to the estimated mean amount per hour.

How much should Drew charge per hour?

£42 p[illegible]our [4 marks]

Answer	£

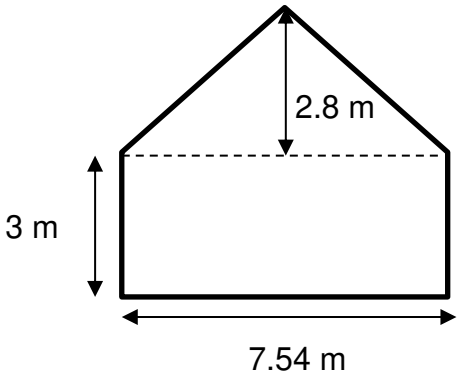
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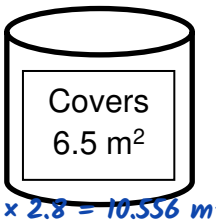
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14. Ashley wants to paint the wall shown:



Not drawn to scale

Ashley finds this tin of paint online:



Area: Triangle = 0.5 × 7.54 × 2.8 = 10.556 m² | Rectangle = 3 × 7.54 = 22.62 m²

Total Area = 33.176 m². Tins = 33.176 ÷ 6.5 = 5.104 tins -> Rounds up to 6 tins

How many tins of paint will Ashley need to buy to paint the wall?

Show your working.

6 tins ✓ [5 marks]

Answer	**tins**

[5]

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15. Quinn wants to buy some rope lights to put around a 164 foot section of a garden.

Quinn finds these rope lights online:

Rope light	A	B	C	D	E
Price per metre	£0.60	£1.99	£1.89	£0.75	£0.50

Quinn wants to buy rope lights that are 8% longer than the section of the garden.

Quinn buys the rope lights with the median price.

How much will Quinn pay in total?

Use 1 foot = 0.305 metres

Median price = £0.75. Length = 164 × 0.305 = 50.02 m.

Net cost = 50.02 × £0.75 = £37.515. Total with 8% = £37.515 × 1.08 = £40.52

£40.5 [6 marks]

Answer	£

[6]

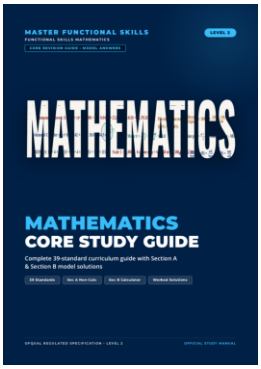
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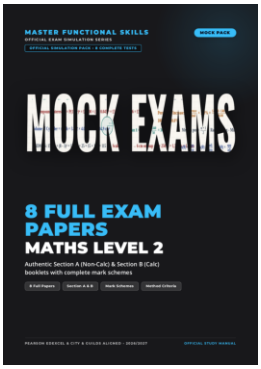


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